DESIGN COMPUTED · FREE GUIDE

Few pieces, strict rules: how muqarnas close a square into a dome

The free guide to the reel on the muqarnas of the Shah Mosque in Isfahan: what they are, how tiers of small cells span a square room, where the idea comes from, and a paper exercise to build one.



What muqarnas are

Muqarnas is a three-dimensional form made of tiers of small niche-like cells. Each tier projects a little over the tier below, like a row of corbels. Western writers often call it 'stalactites' or 'honeycomb'.

It appears on domes, vaults, entrances (iwans), pendentives and prayer niches. In Iran it is often built in brick, sometimes covered with plaster or glazed tile, as at the entrance of the Shah Mosque in Isfahan (17th century).

Few pieces, strict rules

Each cell is built from a limited set of simple prismatic elements, combined under precise rules inside a geometric framework with a few axes of symmetry. That is why a surface with row after row of cells can still be drawn and built from a small kit of parts.

Muqarnas also appear in plans: a medieval architect's drawing of two muqarnas vaults survives on the Topkapı Scroll, a pattern book compiled possibly in the late 15th or 16th century.

How it closes a square into a dome

Muqarnas evolved from the squinch, the small arch that lets a round dome sit on a square room. Tier after tier, each cell steps out a little further than the one below, until the opening that was wide at the bottom is small enough to be closed at the top.

So it is not only decoration: it is also a way to move from a square plan to a round dome.

Built examples

Imam Dur Mausoleum, Samarra (1090)	The oldest surviving full muqarnas dome.
Qarawiyyin Mosque, Fez (1134-1143)	The earliest complete muqarnas vaults in the western Islamic world.
Sala de Dos Hermanas, Alhambra (14th century)	At least 5,000 cells and sixteen small domes around the edge.

Shah Mosque, Isfahan (17th century)	Muqarnas in the entrance iwan, covered in glazed tile.

Try it yourself: build a paper muqarnas

1. Cut three strips of card, each 2 cm wider than the one before, and fold each strip into a row of small arches (cells).
2. Glue the narrowest row to a wall of card, the next row above it, and so on, so each row sticks out a little more than the one below.
3. Do the same from the other side of a 20 cm gap. Watch the gap close as you add rows.
4. Count how many different cell shapes you used. The goal is as few as possible.
5. Colour one row differently and look at the pattern it makes. You have used the rule 'few pieces, strict rules'.

The rule to remember

Before inventing new parts, ask what a small set of simple parts could do if you repeat it under strict rules.

Default answer	A unique piece for every spot.
Form-finder answer	A short list of pieces and the rules for combining them.
Check	Ask: how many different parts does this really need?

Go further

Muqarnas (Wikipedia)
https://en.wikipedia.org/wiki/Muqarnas

Shah Mosque, Isfahan (Wikipedia)
https://en.wikipedia.org/wiki/Shah_Mosque

Topkapı Scroll (Wikipedia)
https://en.wikipedia.org/wiki/Topkap%C4%B1_Scroll

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Facts: Wikipedia (Muqarnas), Topkapı Scroll.

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